ArgOS: a resource governor for a re-configurable vessel

Team entry for Fincantieri's **Software Defined Vessel** challenge (Maritime Dual-Use Hackathon, San Francisco, 16-18 Oct 2026). ArgOS, after the Argo, is the operating system the ship does not have: it owns the hull's finite physical budget and decides, in seconds and in one plain sentence, which payloads come aboard, where they go, who wins when two want the same thing, and what is shed when a generator trips.

> **Status (Sat 14:45 PDT): D1 to D4, the decision records and the state estimator are built and reproduce every oracle case; the tick loop and the screen are next.** Everything marked *planned* below is design.

Run it

Works today (Python 3.12 via uv). The tests check the cards and priority files against their schemas, the data against the brief's arithmetic, the expected outcomes and the Scenario B timeline against the cards, and every requirement link:

```
uv sync
uv run pytest
uv run sdv admit G4 --installed G1,G2,G3 --mission survey
uv run sdv admit G4 --installed G1,G2,G3 --set G4.hangar.mass_t=0.8
uv run sdv run --mission sar --log-dir runs/b-sar && uv run sdv replay runs/b-sar --mission sar
```

`sdv admit` runs D1 and D2 on the real engine: verdict, binding line, every ledger line, placement, ranked yes-if options and the one-genset-down check, with the time taken against the 30 s ceiling. `sdv run` plays Scenario B through the tick loop (simulator, estimator, decisions) and prints each decision's sentence; `sdv replay` re-runs the logs without the simulator and checks every snapshot and record byte for byte.

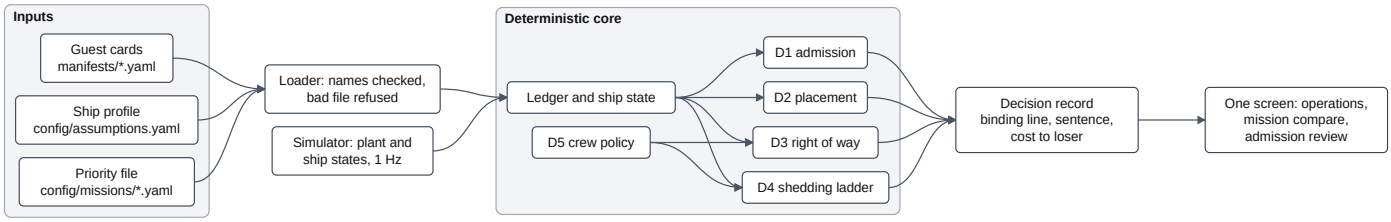
The one demo command (planned; it needs the real engine, which the brief forbids mocking) starts the decision engine and the screen and loads Scenario B; the priority file is picked and swapped on screen. Entry point `sdv.cli:main` (`src/sdv/cli.py`; `demo` planned):

```
uv run sdv demo
```

The idea in one line

One ledger, one manifest language. Everything that uses or supplies a resource is described the same way and counted in one ledger of margins across the brief's eight resource classes: power, heat rejection, deck positions, mast positions, mass and stability, data links, compute and response time. Every decision names the ledger line that bound it. Host commitments are never shed; guests are admitted, placed, throttled or shed; supply can degrade; crew are both hands and authority. Details: GAMEPLAN, Core model.

Architecture



The inputs exist and are tested; the rest of this section is planned. Card conditions are typed: every signal, position and capability name is checked against the ship profile, and an unknown name refuses the file. Faults enter only as debounced, confirmed plant signals: a genset trip on 2 of 3 signals, a

cooling fault on the plant alarm plus flow or delta-T; each record keeps the raw-event and confirmation times. A stale feed falls to its conservative value, and the sentence says so. Safety commands (stop, hold, secure; 200 ms, A.8) are answered from a table precomputed every tick, never by a solver. The acoustic link to the seabed vehicle is up only on a steady course, in quiet, with no deck machinery running, and is assumed lost the moment one of those fails. A replay reproduces every record field except wall-clock latency and timestamps, which go to a separate timing log.

The screen is one page with three views: Operations (schematic, eight gauges, decision log, with the constraint ledger as its drill-down), Mission compare (two runs, one per priority file, on one timeline) and Admission review (the jury's card). An Arduino VENTUNO Q board (measured) is an instrument only: it runs the same container and reports watts, latency per decision class and CPU share against A.7 for the Physical Account; every demo input comes from the laptop. Data contracts: docs/ARCHITECTURE.md.

Prior art. Ships already shed load: a power management system trips non-essential consumers in a fixed preferential-trip order when a generator is lost. ArgOS is a mission-aware, explainable version of that ladder: the order follows the active mission, throttling comes before tripping, and every step names the line that bound it.

Decision classes

D1 to D4 read one ledger (`src/sdv/ledger.py`) and write the same decision record; ceilings are the brief's (A.8).

Class (ceiling)	Component	What it optimises
D1 Admission (30 s)	`src/sdv/admission.py`	A verdict that holds at both gensets, 200 kW and 30 °C, and survives one genset down by throttling (rule below); for yes-if, the least mission value lost
D2 Placement (30 s)	`src/sdv/placement.py`	No installed part left without a position; most parts placed; fewest port moves; preferred positions first
D3 Right of way (2 s)	`src/sdv/arbitration.py`	Hard limits first (the drone's 12-minute landing rule), then mission-weighted value from the priority file; cost to the loser stated
D4 Degradation (5 s, 60 s)	`src/sdv/shedding.py`	Vessel-critical never shed, hard limits first; throttle before shed; then the profile's criticality tier (discretionary first, mission-critical last), its `shed_order`, then least mission loss per kW freed on the binding line, rechecked each step; ladder precomputed every tick
D5 Crew as a setting	`src/sdv/crew_policy.py` (planned)	Per rule and crew level (14, 6, 0): confirm, standing plan, automatic or forbidden; a priority file may override a level, never to confirm at 0 (A.6); rule R3 in the Scenario C memo
D6 Physical Account	`tools/physical_account.py` (built)	Every assumed number, with its reason and where code uses it: docs/PHYSICAL_ACCOUNT.md

D1 rule. Judged at both gensets, the 200 kW reserve (A.1) and cooling at 30 °C (A.10); it must also survive one genset down by throttling to the 95 kW gate, otherwise yes-if, with the raw one-genset-down line always shown (DEC-35). In port, installed guests may move between bays, each move reported with its crane time (DEC-37). Links are judged at guests' minimum rates, any nominal shortfall shown as degraded quality (DEC-38). The seawater tipping point is reported. D1 and D2 search by enumeration: six guests and six positions, Bay 1 in halves, give a few thousand assignments, so the binding line and the rejected alternatives come out directly, in milliseconds.

Deliverables

- **Prototype, one documented command:** `uv run sdv demo` (Run it); planned.
- **Manifest schema, five cards plus one of ours:** manifests/schema.json, G1-G6 (G6 is our desalination unit), how to add a card; validated by tests.
- **README (at most 3 pages), Physical Account (1 page), Scenario C memo (PDF, at most 2 pages):** this file, the account (from config/assumptions.yaml and the cards) and the memo;

`tools/build_pdfs.py` builds each PDF and checks its limit.

- **Live demo, 90 s of Scenario B, then Scenario A with the jury's card:** storyboard, planned. Scenario B runs G3, G4 and G5 (DEC-32). Minute 8: D3 decides between the drone (G4) and the seabed vehicle (G3) under both priority files. Minute 12: a genset trips; the load (48 kW survey, 81 kW rescue) holds under the 95 kW ceiling, and D4 shows both armed ladders, each starting with the comms node (G5) to low power, with no step needed. Minute 19: a labelled, injected pump fault makes cooling bind (81 against 75 kW); D4 slows G4's charge to 10 kW (survey file) or takes G5 to 8 kW (rescue file) (DEC-40).

Repository map

Every path and what it holds: docs/REPOSITORY_MAP.md. Every requirement, and how we address it: docs/REQUIREMENTS.md. Decisions and who made them: WORKLOG.md.

Limitations (today)

- The tick loop and the screen are not built yet. The simulator is Siddharth's; until it lands, the estimator (docs/STATE_ESTIMATION.md) is tested against a labelled stub of it.
- Numbers come from the brief's annexes and our labelled assumptions, never from a real vessel.
- By choice, both power lines survive a genset trip (A.9); a 40 % cooling loss is information only (DEC-36).
- With nobody aboard, rule R3 trusts a deck latched clear at the last crew sweep; nothing aboard sees loose gear shift after that sweep.

With another week

- Calibrate the simulator's plant and sea models against the public datasets already identified.
- Measure ArgOS under load on hardware closer to A.7's rack, and scale the Physical Account's figures.
- Settle the emission-silence rule with G4 airborne as one D3 rule and show it live.
- Work through more priority files and jury cards, including hostile ones.

Declarations and licence

AI and software, with licences and origins: docs/SOFTWARE_LOG.md. Claude (Anthropic) is the coding and research assistant; no model from a China-based company is used. Only open-source third-party code or our own; code licence MIT. Data (planned): sea temperature, waves and wind from Open-Meteo (a source the brief names), cached offline. Weather data by Open-Meteo.com, CC BY 4.0.